ICT Express 11 (2025) 754–768

Contents lists available at ScienceDirect

ICT Express

journal homepage: www.elsevier.com/locate/icte

Federated learning and TinyML on IoT edge devices: Challenges, advances, and future directions

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ARTICLE INFO

Keywords:
Federated learning
TinyML
IoT
Edge devices
Resource constraints

ABSTRACT

This paper examines the integration of Federated Learning (FL), TinyML, and IoT in resource-constrained edge devices, highlighting key challenges and opportunities. It reviews FL and TinyML frameworks with a focus on communication, privacy, accuracy, efficiency, and memory constraints. We propose a novel FL-IoT framework that combines over-the-air (OTA) AI model updates, LoRa-based distributed communication, and lossless data compression techniques such as Run-Length Encoding (RLE), Huffman coding, and LZW to reduce transmission cost, optimize local processing, and maintain data privacy. The framework features Raspberry Pi-based aggregation nodes and microcontroller-based IoT clients, enabling scalable, low-power learning across heterogeneous devices. Evaluation includes memory usage, communication cost, energy consumption, and accuracy trade-offs across multiple FL scenarios. Results show improved scalability and significant power savings compared to baseline FL setups. The proposed framework is particularly impactful in applications such as smart agriculture, healthcare, and smart cities. Future directions for real-time, privacy-preserving edge intelligence are discussed.

1. Introduction

The technological landscape of the 21st century is increasingly defined by the convergence of various advanced fields - among them, the Internet of Things (IoT), Machine Learning (ML), and Federated Learning (FL). The integration of these domains is not merely a technological evolution but a revolution in how data is processed, analyzed, and utilized, especially with the emergence of Tiny Machine Learning (TinyML) and its application to IoT edge devices. This comprehensive review aims to delve into the intricate world of Federated Learning, focusing on its integration with TinyML and IoT edge devices, exploring the potential, challenges, and the roadmap ahead [1]. The concept of IoT has matured from a theoretical idea to a tangible reality that permeates everyday life. IoT refers to the vast network of interconnected devices that communicate and exchange data. These devices range from simple sensors in home automation systems to complex industrial machines. The proliferation of IoT has been exponential, with billions of devices now connected, each contributing to an ever-growing repository of data. The International Data Corporation (IDC) predicts that by 2025, there will be over 41.5 billion IoT devices, generating around 79.5 zettabytes of data [2]. This data, while a treasure trove of information, presents significant challenges in terms of processing, privacy, and security.

Machine Learning has been instrumental in harnessing the potential of IoT [3,4]. By applying ML algorithms to IoT data, devices can make intelligent decisions, predict outcomes, and automate processes without human intervention [5]. However, traditional ML approaches often rely on cloud computing, where data from various IoT devices is sent to centralized servers for processing. This model, while powerful, faces challenges such as latency, bandwidth limitations, and privacy concerns [6,7] TinyML emerges as a solution to these challenges. It represents a field of study that focuses on implementing machine learning models on low-power, resource-constrained devices like those used in the IoT [8]. TinyML enables on-device data processing, thereby reducing the need for constant data transmission to the cloud, enhancing privacy and efficiency [9]. The integration of ML directly into IoT devices facilitates real-time analytics and decision-making at the edge, where data is generated [10]. Moreover, TinyML represents a significant transformation in the field of artificial intelligence. This approach brings AI

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https://doi.org/10.1016/j.icte.2025.06.008
Received 20 February 2025; Received in revised form 30 April 2025; Accepted 10 June 2025
Available online 19 June 2025
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capabilities directly to the edge, utilizing small scale devices like microcontrollers to run AI algorithms [11]. TinyML offers the advantages of reduced latency, minimal power consumption, and limited bandwidth usage.

In contemporary computing technologies, typical central processing units (CPUs) generally consume up to 90 W, while standard graphics processing units (GPUs) use between 200 and 600 W [12]. In stark contrast, a common microcontroller operates with significantly lower energy, consuming power in the range of milliwatts or microwatts, which is a thousand times less than standard CPUs and GPUs. This remarkably low power consumption characteristic of TinyML devices allows them to function without being plugged in for extended periods, ranging from weeks to even years [13].

In the context of IoT networks, scalability is crucial because these networks often involve vast numbers of devices that need to function seamlessly as the network expands. It refers to the ability of a system, network, or process to handle a growing amount of work or its potential to be enlarged to accommodate growth. As IoT networks grow, the challenge of managing large volumes of data while maintaining performance, efficiency, and security becomes more complex. The traditional approach of centralized data collection and processing can become inefficient and impractical due to increased data transmission, higher latency, and greater privacy risks.

The great scalability of contemporary IoT networks and growing concerns about data privacy may make it impractical for AI techniques that traditionally require centralized data collecting and processing in practical application settings. Federated Learning, a novel approach to machine learning, further enhances the capabilities of IoT and TinyML. FL allows for the training of machine learning models across multiple decentralized devices while keeping the data localized [14]. This approach addresses several critical challenges of traditional ML, including privacy concerns and the inefficiencies of transmitting large volumes of data to centralized servers. In FL, IoT devices can collaboratively learn a shared prediction model while keeping all the training data on the device itself [15]. This decentralized approach not only ensures privacy and security but also opens up new avenues for real-time, on-device learning and decision-making.

In numerous Federated Learning (FL) applications, particularly those in the IoT domain, the client devices often have limited processing capabilities. These devices include microcontrollers and microprocessors, which are commonly found in Unmanned Aerial Vehicles (UAVs), smart home appliances, smart greenhouse systems, vehicle tracking systems and smart wearables. Unlike most servers that are outfitted with advanced Tensor Processing Units (TPUs) or Graphics Processing Units (GPUs), these client devices typically possess less powerful processing units. Moreover, these units are frequently engaged in tasks other than model training. Consequently, the development of effective solutions to address these limitations is crucial for FL to potentially supplant traditional centralized training methods.

The synergy between Federated Learning (FL) and Tiny Machine Learning (TinyML) represents a transformative approach in the realm of AI and IoT, offering a paradigm that is both efficient and privacy-centric [16]. Federated Learning, with its decentralized model training, allows for the aggregation of insights from numerous devices without the need to transmit vast amounts of raw data to central servers. This method aligns perfectly with the ethos of TinyML, which is all about embedding lightweight AI algorithms into small, power-efficient devices at the edge of the network [17]. By combining FL's capacity for collaborative learning across multiple devices with TinyML's ability to perform AI tasks on low-power microcontrollers, this synergy unlocks new potential in smart technology. It enables smarter, more responsive IoT environments, where decisions are made faster and more locally, while also addressing key concerns such as data privacy and bandwidth limitations [18]. This powerful collaboration paves the way for advanced, yet sustainable AI applications in everyday devices, revolutionizing how technology interacts with and learns from the real world in a decentralized manner [19]. While the integration of Federated Learning with TinyML and IoT presents significant opportunities, it is not without challenges. These include technical hurdles such as ensuring model efficiency on resource-constrained devices, addressing communication overheads in distributed learning, and maintaining data security and user privacy [20].

Furthermore, the heterogeneity of IoT devices in terms of computational power and data formats adds complexity to the implementation of FL and TinyML [21]. Looking forward, the continued evolution of this integration promises to not only enhance the capabilities of IoT devices but also to revolutionize how data is processed and utilized in various sectors. Innovations in hardware and software are continuously improving the feasibility and efficiency of deploying advanced ML models on edge devices. As these technologies mature, they are poised to unlock new possibilities in smart cities, personalized healthcare, intelligent transportation systems, and more [22].

The convergence of Federated Learning and TinyML in the realm of IoT edge devices represents a significant stride in the field of technology. It is a step towards a future where data processing is more secure, efficient, and aligned with the needs of an increasingly connected world. As this field continues to grow and evolve, it holds the promise of transforming not just individual industries but the very fabric of our daily lives, making intelligent, decentralized, and privacy-preserving computing a reality. While FL and TinyML have been the subject of numerous surveys, these two subjects are typically researched independently of one another without any integration. Regarding FL, the study in [23] offers an overview of FL's principles as well as how it integrates with industrial IoT, covering privacy and security. While the authors of [24] offer a thorough analysis of the FL concept and IoT, they also emphasize the significance of integrating FL and IoT by offering several significant applications. The authors of [25,26] provided a comprehensive assessment that discusses FL's capabilities and use in smart cities while outlining the main obstacles facing FL in smart city applications. The authors of [27] concentrated on the primary limitations and constraints of FL as well as its practical applications. In the framework of FL, the authors of [28] provided an extensive and methodical analysis of the problem of communication and computation efficiency. Another fascinating study, [29], focused on edge devices with constrained computational resources while presenting FL techniques. In [30], the authors provided a thorough analysis of current FL algorithms that could be used in a medical IoT setting. The intrusion detection systems based on FL are evaluated by the authors in [31] utilizing multiclass classifiers that take into account various distributions for the detection of various IoT assaults. The authors of [32] provide an extensive overview of the difficulties in implementing distributed machine learning models for IoT devices with limited resources, including microcontrollers.

A new classification of FL based on major technical obstacles like privacy and security issues was presented by the study in [33]. A FL survey on non-IID data is offered [34], Non-IID data refers to situations where data distributions vary across clients, meaning that each device may observe different types, frequencies, or patterns in its local data. In both horizontal and vertical FL, this survey provides a detailed investigation of the impact of non-IID data on parametric and non-parametric machine learning models. Nevertheless, several surveys address TinyML. The authors of [35,36] provide a thorough analysis of TinyML that includes frameworks and techniques for integrating TinyML with network technologies like 5 G. Other researchers, including those in [37–39], concentrated on hardware systems that supported TinyML and compared microcontrollers with ML support. Furthermore, most other research [37,40–43] offer several TinyML applications, including those for smart farming, healthcare, the environment, and anomaly detection. Table 1 summarizes a comparison study of previous surveys about FL and TinyML.

As has been highlighted in Table 1, no literature covers the aspect of fully integrating FL, TinyML, and IoT technologies. Thus, there is a need

Table 1
Focus of discussion on the Existing surveys of Federated Learning and TinyML.

Ref.	Application	Integration of FL with TinyML and IoT	Focus on the Discussion in these surveys					
			communication	privacy and security issues	Frameworks, Memory and processing	Accuracy and efficiency	Scalability and cost	Latency and IoT edge devices
[23], 2023	Federated Learning in IIoT	Yes	No	Yes	Yes	No	No	No
[27], 2024	FL concepts and application	Yes	No	Yes	Yes	Partial	No	No
[25]	FL application in smart cities	Yes	Yes	No	Partial	No	No	Yes
[24]	FL concepts and application	Yes	Partial	No	Yes	Partial	No	No
[26]	FL in Smart Cities	Yes	No	Yes	Yes	No	No	Yes
[44]	FL Concepts	Yes	Yes	Yes	Partial	No	No	Yes
[31]	FL on edge devices	Yes	Partial	No	Yes	No	Yes	Yes
[45]	FL Application	Yes	Yes	Yes	Yes	Partial	Yes	Yes
[46]	TinyML with MCUNet for ImageNet-scale AI on IoT devices.	Yes	No	No	Yes	Yes	No	No
[47]	TinyML with LPWAN and 5 G networks	Yes	Yes	No	Yes	Yes	Yes	Yes
[48]	TinyMLaaS	Yes	Yes	Yes	Yes	Partial	Yes	No
[49]	ML with IoT and Edge machine learning	Yes	No	No	Yes	Yes	Yes	No
[50]	ML/DL on Edge intelligence	Yes	Yes	Yes	Partial	Yes	Yes	Partial

for framework that enables frequent AI model updates on IoT devices, significantly reducing power consumption and communication costs while fostering more autonomous, intelligent microcontroller systems.

The key contributions of this paper are summarized as follows:

- We provide a comprehensive review of integrating FL and TinyML with IoT edge devices, particularly microcontrollers, and explore their transformative impact across various industries.
- We identify and analyze the critical challenges that must be addressed to optimize model efficiency on IoT edge devices.
- The paper highlights significant gaps in existing research, particularly the lack of comprehensive studies and practical applications that fully integrate FL, TinyML, and IoT technologies.
- We propose innovative solutions, including a novel framework, that address these challenges and pave the way for smarter, more efficient microcontroller systems.
- Finally, we suggest future research directions focused on further improving the deployment of AI models on microcontrollers, with an emphasis on overcoming current limitations.

The remainder of this paper is structured as follows: Section 2 provides a detailed introduction to TinyML, IoT edge devices, and the resource limitations inherent in edge computing. Section 3 offers a thorough explanation of FL, covering essential concepts such as model updates, aggregation methods, communication protocols, and edge learning. In Section 4, we present various scenarios where FL empowers edge devices and demonstrate the compatibility of FL with TinyML and IoT systems, while addressing interoperability challenges and potential solutions. Finally, Section 5 concludes by outlining the key challenges and future directions for the integration of TinyML, IoT edge devices, and FL. The structure of this paper is visually outlined in Fig. 1.

2. Federated learning (FL) TinyML and IoT fundamentals

In this chapter, the details on federated learning (FL) and TinyML, types of FL, algorithms and security breaches, have been provided. It shows the possibility of creating an actual prototype by merging FL, TinyML, and IoT technologies. The idea on how the microcontroller is compressed to have a small size data during transmission has been studied.

2.1. Federated learning

Federated Learning (FL) is a machine learning method that runs an algorithm on multiple devices or servers with local data samples without relocating them [51–53]. This technology has proved beneficial in cases where data privacy, data security, and data rights access are problems [54]. FL is an innovative approach based on getting the model to the

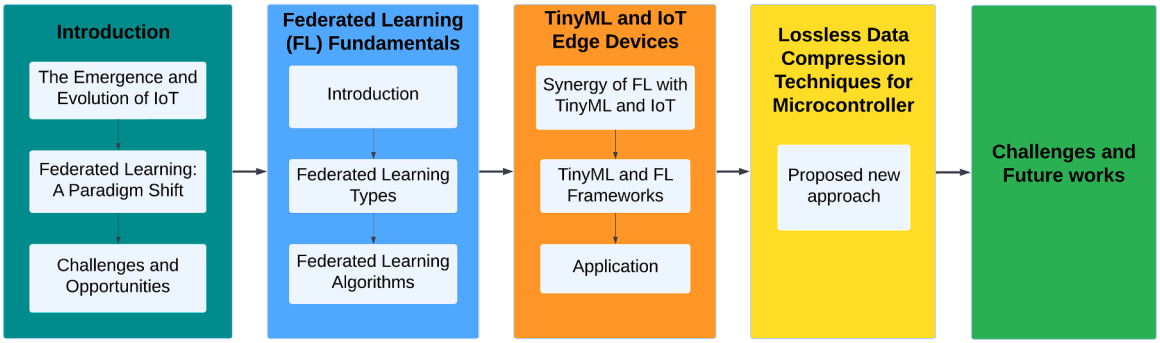


Fig. 1. Organization of this article.

data, opposing the outdated method of bringing data to the model. In this case, it is even more critical for the data models to maintain their confidentiality and efficiency because of the extensive data or its character [55].

The main aim of FL is to implement the company's model, where data privacy and security are not compromised. Under this plan, the global model is copied into each participant's device and trained locally using local data [56]. Next, the device will send the model updates back to the central server (such as weights or gradients) as shown in Fig. 2. The updates are consequently accumulated to upgrade the model's version for the globe. The sending, updating, and accumulating procedures are repeated until the model reaches the highest level of accuracy [57]. The Federated Learning (FL) process can be summarized in three main points:

- **Initialization:** The process is started by selecting a model architecture and providing a central server. The devices selected have a variety of data and computation abilities. The communication protocols and the data privacy measures are set up [58].
- **Local Model Training:** The devices just train local models on their data and do not disclose raw data. These models are built locally and updated iteratively using their regions' data [59].
- **Model Aggregation**: Models are updated by the local models, and the updated models are sent to the central server. These are subsequently integrated to form the global model. It helps to make an interconnected network model that can learn from all the devices and keep the data safe as well. These sequential steps are repeated iteratively to achieve the best possible performance of the global model [60].

FL can be divided in various ways, for instance, based on data sources, communication techniques, network architecture, model size, privacy issues, and incentive participation. In this part of the analysis, we will examine each type of categorization in detail, providing relevant examples. When a user is developing an FL model, the user must be aware of the data distribution by accounting for the features and the special cases.

2.1.1. *Federated learning types*

FL can be divided into three main types based on the distribution of data samples across networks and their feature spaces as shown in Fig. 3: 1) Horizontal FL; 2) Vertical FL; and 3) Transfer FL.

- **Horizontal Federated Learning (HFL)** is a collaborative machine learning technique where multiple organizations or devices with similar data structures (features) collaborate to train a shared model without sharing their actual data. This approach is particularly useful when datasets from different sources have the same feature space but belong to different entities, thus enabling privacy-preserving collaborative learning. Each participant trains a local model on their own data and shares only the model updates (gradients) with a central server, which aggregates these updates to improve the global model [61].

  The global model is the first step of HFL, which is placed on the central server for initialization. The trained model will be shared with different clients (which can be smartphones or IoT devices) for local execution. Every client applies model training on their dataset and computers updates, usually in the form of gradients or weights. These updates are then transmitted to the central server, where they are averaged, mostly by using averaging methods, to improve the global model. This is a cyclic process; the client gets the updated model and then sends it back for further modifications. Then, the cycle continues until the model has met a satisfactory level of performance or convergence [62].One of the critical attributes of HFL is the homogeneous nature of the data across all clients. Though the individual points of data might vary, the general structure of the data or the features that are constant within the dataset remains the same. This kind of uniformity is the basis for the successful training process of the HFL model. Consequently, this strategy benefits privacy preservation since the data samples remain with clients, and only model updates are shared [63].
- **Vertical Federated Learning (VFL):** is another collaborative learning technique where multiple organizations or devices with different feature spaces but overlapping sample spaces work together to train a shared model. In VFL, participants have different attributes of the same set of samples. This technique is beneficial when different entities hold complementary information about the same

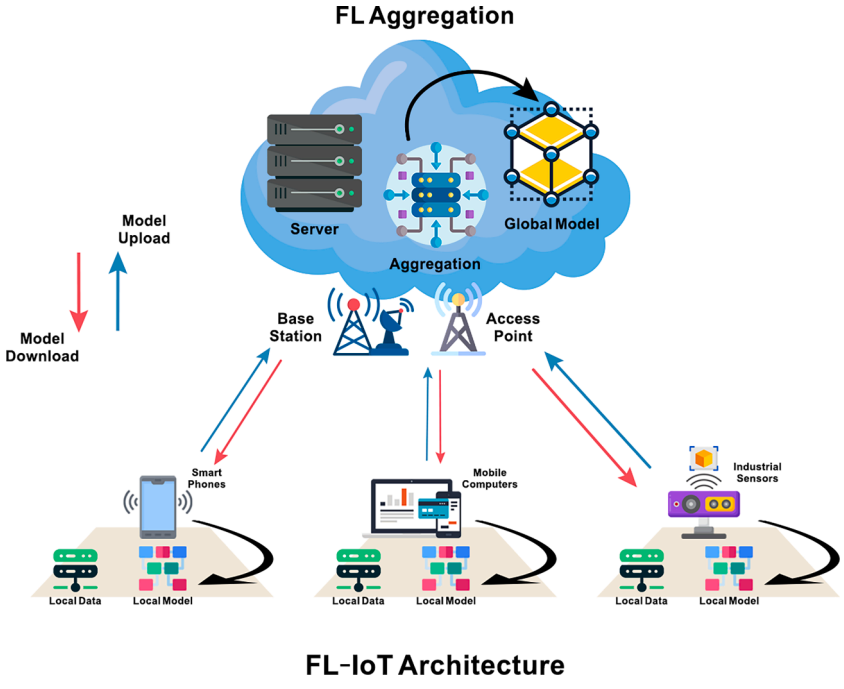


Fig. 2. The network architecture and communication process for FL-IoT.

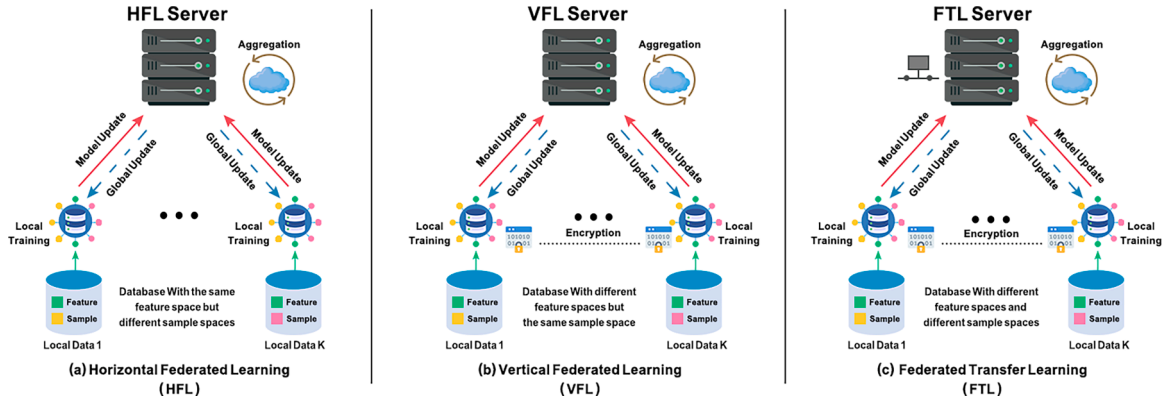


Fig. 3. Types of FL models with data partitioning: (a) Horizontal Federated Learning, (b) Vertical Federated Learning, (c) Federated Transfer Learning.

set of entities or individuals.

The collection of such data is an essential part of constructing a more responsive traffic management system and a cleaner environment in the VFL system, where different entities present different features of the same dataset. In contrast to the HFL, where entities are non-homogeneous with a similar feature set, VFL focuses on combining the same sample set but with different feature sets [64]. For instance, the two companies can have different demographic data, and the other one can own the transaction data of the same customer set. VFL offers a venue to create models that jointly cover all the features. They do not have to share their raw data directly. This strategy is essential in cases where data can't be shared for privacy reasons or competitive borders.

The first step of VFL is data alignment, where participants align their datasets using the same identifiers, ensuring they analyze the same samples [65]. In essence, vertical federated learning is a critical solution for collaborative data analysis in situations where data cannot be shared directly. Its ability to merge the different data types from multiple sources to protect privacy and security signifies a new milestone in machine learning and data science.

- **Federated Transfer Learning (FTL)** Transfer Learning is a machine learning technique where a pre-trained model developed for a specific task is reused as the starting point for a model on a different but related task [66]. Instead of training a model from scratch, Transfer Learning leverages the knowledge learned from the pre-trained model to improve performance and speed up the training process of the new task. This approach is particularly beneficial when there is limited labeled data available for the target task [67]. This approach is helpful in optimizing machine learning models, particularly in cases of absence or the lack of variety in the domain.

In the FTL, a process involves a pre-trained model trained on a source data set, who may have already learned some useful features or representations for a different but related task. This model modification is then integrated into federated learning architecture [68]. For instance, a model trained on a big, centralized dataset of medical images (source domain) could be employed to advance the diagnostic algorithms of a network of hospitals (target domain) with its own local dataset. FTL's central advantage is that the data labeled in the target domain is unnecessary, which is a common problem in machine learning projects [69].

On the other hand, the FTL has specific problems [70], primarily due to the heterogeneity of data and the dissimilarity in feature distribution across the source and the target domains. The application of knowledge transfer requires the development of relevance and benefit tuning and adaptation strategies. In addition, the same data privacy and security issues during the transfer learning phase should be considered in the case of standard federated learning, for example, when dealing with sensitive information such as medical or financial data [71].

2.1.2. *Federated learning algorithms*

Federated Learning (FL) algorithms empower machine learning across decentralized devices [72–74], prioritizing data privacy and system efficiency. Many versions of the algorithms have been updated, each of which is meant to work well in a particular situation. This section highlights a few popular and useful FL algorithms to encourage researchers to release a better FL version.

- Federated Averaging (FedAvg) [75–78] is a cornerstone algorithm in Federated Learning, introduced by McMahan et al. It's recognized for its straightforward approach, where clients independently train models on their local data and send these model updates, typically the weights, to a central server. The server's role is to average these updates, essentially synthesizing a global model that it then redistributes to the clients for further refinement. This process significantly reduces the communication overhead compared to transmitting raw data or updating after every local training epoch. However, FedAvg makes an implicit assumption that the data across clients are identically distributed, which is often violated in real-world scenarios. This leads to potential performance degradation, especially in the presence of heterogeneous data distributions and systems [79]. Different researchers working on enhancing FedAvg, the authors in [80], present an updated version of FedAvg called N-FedAvg, which changes the random selection of the clients in traditional FL to sequential selection of clients and makes the learning rate changed dynamically. The results showed that the N-FedAvg increased the accuracy by 1.34 %, But it still needed more evaluation on different datasets and different applications.
- Federated Stochastic Gradient Descent (FedSGD) is akin to FedAvg. Still, it is more aligned with the traditional Stochastic Gradient Descent (SGD) technique, adapted for the federated setting. It differentiates itself by the frequency of updates; rather than waiting for a full local training epoch to complete, updates are communicated for each training example or small batch. This can lead to more accurate and faster convergence in specific scenarios due to the frequent updates. However, the increased communication demand makes FedSGD less practical in environments where bandwidth is limited or costly [81].
- Federated Proximal (FedProx) emerges as an enhancement to FedAvg, particularly addressing the challenges arising from the heterogeneity in client data distribution and system capabilities. Its novel approach introduces a proximal term to the local optimization problem, effectively anchoring the local updates closer to the global model. This robustness to non-Independent and Identically Distributed data and system heterogeneity comes at a cost, as the proximal term introduces additional complexity in hyperparameter tuning, necessitating a delicate balance with the learning rate for optimal performance [82].

Secure Aggregation (SecAgg) is a very interesting and powerful

approach. It is a safe, private, and cryptographic approach to securely aggregate model updates that are done by the participants in the network. The main idea is to make sure that the central server, or any hostile actor, possesses only the aggregated model update without having the ability to uncover individual contributions. The privacy level of SecAgg has undoubtedly increased, but at the same time, the framework introduces computational and communication overheads because of the cryptographic operations involved. Therefore, the implementation of the framework should be handled with care to avoid any vulnerabilities [83].

- The more recent and dynamic approach known as Adaptive Federated Optimization (AFO) represents a way of adaptation. It's unique in its adaptability, whereby the learning rates for the server and the clients are adjusted automatically to be more suitable for the differences in the client updates and data distributions. This adaptiveness can be a critical factor in model performance and speed up convergence. On the other hand, the adaptive control elements are complex to implement and tune. The necessity for the rigorous analysis of the stability and convergence of the control system cannot be underestimated [84].

Table 2 provides a comparative overview of key Federated Learning (FL) algorithms, highlighting their primary features, strengths, and challenges. The table includes widely used algorithms such as FedAvg, which reduces communication overhead by averaging model updates, and FedProx, which improves robustness in heterogeneous environments through a proximal term. More advanced techniques like Secure Aggregation (SecAgg) focus on enhancing privacy through cryptographic methods, while Adaptive Federated Optimization (AFO) dynamically adjusts learning rates to optimize model performance across diverse client data distributions.

3. TinyML and IoT edge devices

TinyML, a new development in the field of Machine Learning on Tiny Devices (MTD), combines the immense power of Machine Learning (ML) with the small form factor and low power consumption of embedded systems [85,86]. For low-power MCUs and devices, which are often imperfect in memory, giving out power and energy possessions, the ML models can be run by TinyML. This new net harvests the conditions whereby the gadget can conduct complex data giving out without the requirement to eternally attach to the cloud or other servers. This is because the device can act and make choices faster without the need for the internet [87].

One of the major benefits of TinyMLs is that it can use AI in small devices, such as sensors, smartwatches, and so on, to track health and agriculture equipment [86]. This functionality, therefore, comes in handy in fields where data processing and low power consumption are critical, including data analysis, artificial intelligence, and the internet of things. For instance, TinyML can be applied to study wildlife in conservation projects [88], do gesture recognition in consumer devices [89], and perform predictive maintenance [90] in industrial environments, all of which will protect privacy and lower latency and bandwidth consumption.

IoT edge devices are a crucial component of the IoT ecosystem. These devices consist of microcontrollers, sensors, and actuators which are located at the edge of the network and perform data processing tasks locally. By processing data at the edge, these devices reduce the need to transmit large volumes of data to central servers or the cloud, which can help reduce latency and bandwidth usage, as well as improve response times.

While TinyML and IoT edge devices both contribute to empowering IoT systems by enabling local data processing, they serve different purposes and have distinct characteristics. TinyML focuses on bringing machine learning capabilities to highly resource-constrained devices, while IoT edge devices encompass a broader range of functionalities and hardware capabilities.

The integration of Federated Learning (FL) and TinyML in IoT edge devices enables efficient, decentralized data processing, as illustrated in Fig. 4. The process starts with Data Collection, where data is gathered locally from IoT devices [89] Instead of sending raw data to a centralized server, FL allows devices to collaborate in the FL Model Training phase, where machine learning models are trained locally without compromising sensitive data. This decentralized approach ensures data privacy while utilizing distributed data across devices. After training, the model is deployed on edge devices during the TinyML Local Processing phase, enabling real-time, on-device data processing. In the final step, Real-Time Decisions are made based on the processed data, such as detecting anomalies or generating predictions. This process minimizes latency, reduces bandwidth usage, and enhances privacy, making it ideal for resource-constrained IoT environments.

The combination of FL, TinyML, and IoT is enabling a new and improved approach to machine learning. Due to this convergence, many IoT devices will be able to take advantage of the distributed processing power to learn from local data and keep sensitive information at home. In addition to this, Edge AI is the state of art of using AI technology for localized data processing, which is carried out at the network edge where the data are created rather than being processed in remote data centers or the cloud. The integration of Tiny Machine Learning (TinyML) and Federated Learning (FL), which are the fundamental components of the Internet of Things (IoT) infrastructure, is the basis of the new paradigmatic change. Furthermore, data processing locally on the Edge AI devices makes a faster response time for the AI-powered applications. This is critical for realtime applications requiring millisecond choices, such as industrial automation, healthcare monitoring systems, and driverless cars. Furthermore, given the exponential proliferation of IoT devices, transferring raw data from many IoT devices across the network to a central server might cause congestion and require a large amount of bandwidth. By use of localized data processing, Edge AI significantly minimizes the amount of data that requires transmission [91].

Privacy issues are crucial with the growing number of IoT devices in delicate areas, including homes, healthcare, and personal electronics. Edge AI allays these worries by guaranteeing that data is processed locally and that only crucial information or model updates are sent across the network. Federated Learning, as discussed in [92], plays a pivotal role here by allowing AI models to be trained collaboratively without the need to share raw data, thus preserving user privacy. Without relying on cloud computing, Edge AI enables devices to make decisions based on local data insights [93]. This feature is especially helpful when internet connectivity is spotty or nonexistent. With IoT ecosystems' increasing complexity and size, the centralized processing approach is becoming less and less viable. With its decentralized

Table 2
Comparison of key FL algorithms: features, strengths, and challenges.

Algorithm	Structure of FL	Ensuring Privacy and Security	Communication Overhead	Ensuring Adaptability	Efficiency
FedAvg	Number of nodes, Training model	Yes	No	No	High
FedSGD	Continuous updates per batch	No	High	Yes	Moderate
FedProx	Proximal term for heterogeneity	Yes	Moderate	Yes	High
SecAgg	Secure aggregation protocols	Yes	High	No	Moderate
Adaptive FedOpt	Adaptive learning rates	Yes	Moderate	Yes	High

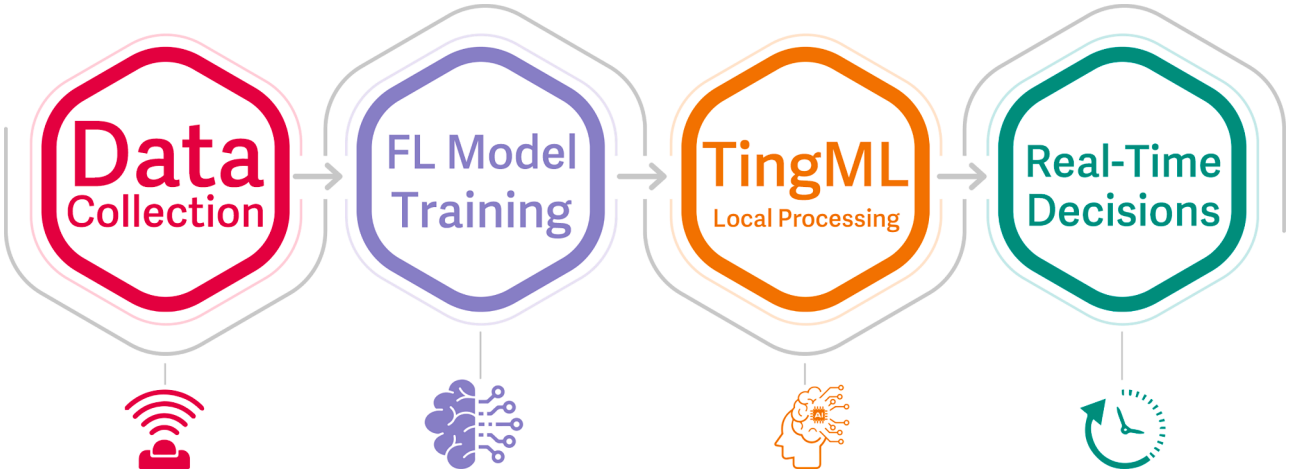


Fig. 4. Data processing workflow for FL and TinyML in IoT edge devices.

approach to AI model training and execution, Edge AI provides a scalable solution that improves system performance [94].

A few researchers are currently engaged in developing and applying TinyML and federated learning applications, The first open-source version of FL designed specifically for the IoT's most resource-constrained devices, such as microcontrollers (MCUs) and devices with small CPUs, TinyFedTL is introduced by the authors in [95], marking a significant milestone in this field. They effectively show that privacy-centric FL is both practical and efficient on devices with low memory capacities (less than 1 MB) by using TL as a fundamental task. This discovery highlights the possibility of more extensive FL implementation on various devices, expanding the opportunities for machine learning application in the IoT. In [96], the authors investigate how FL is improved by TL on a variety of IoT devices, such as PCs, ESP modules, Arduino, and Raspberry Pi. Their method maintains data privacy while achieving impressive results in classification and regression tasks. It also allows on-board machine learning training. The paper [96] compares the system's effectiveness to conventional TinyML models. It discusses the difficulties associated with node integration and unbalanced datasets. Even so, the direct utilization of real-time data training on the microcontroller results in significant power consumption, hence rendering it unsuitable for battery-powered operation. This is considered the most significant drawback of this method of operation

A keyword spotting application that pioneers on-device training with federated learning on Arduino Nano 33 BLE Sense boards is also explored in [97] .Their experiments reveal the performance problems of integrating model training with federated learning on modern microcontroller platforms and on-device training. In supervised learning, users verbally provide data and label for the KWS model via push-button interactions. The research shows that on-device federated learning for KWS is feasible, but it also finds performance obstacles, particularly in processor speed and memory constraints, indicating the need for further optimization and efficiency improvements. Additional researchers attempt to use FL over the LoRa network. In [98], the authors investigate the cutting-edge idea of embedded federated learning (FL) on microcontroller boards, utilizing communication over the LoRa mesh network. They use a dual-board strategy, combining a TTGO LORA32 board for FL networking with an Arduino Portenta H7 board for machine-learning activities. This configuration is evaluated on a keyword spotting job, proving the system's viability and presenting a distributed, re-trainable application running at the small edge. Though scalability, resilience to network outages, and energy consumption optimizations are left as topics for future study, the report mainly concentrates on the proof of concept and preliminary performance analysis.

4. TinyML and FL frameworks

The growing demand for Tiny Machine Learning (TinyML) solutions has led to the development of various tools and frameworks designed to deploy machine learning models on resource-constrained devices. These frameworks are tailored to meet the unique challenges posed by edge computing, such as limited processing power, memory, and energy efficiency, while enabling real-time inference and decentralized data processing.

TensorFlow Lite, developed by the Google Brain Team in 2017, remains one of the most versatile and widely adopted frameworks for edge applications [99]. Its ability to efficiently handle image recognition, audio classification, and object tracking on microcontrollers has made it a go-to solution for developers looking to deploy machine learning models in IoT systems. Similarly, uTensor from ARM has been optimized for low-power devices, focusing on applications such as gesture recognition and audio classification, which are particularly relevant in wearable technologies and energy-constrained environments.

As shown in Table 3, several other tools have also made significant contributions to the field. Edge Impulse, launched in 2019, stands out for its ease of integration, rapid prototyping, and suitability for industrial IoT applications such as asset tracking and predictive maintenance. Its real-time data processing capability on edge devices is critical for industries where latency and power efficiency are top priorities.

For more specialized industrial settings, NanoEdge AI Studio focuses on anomaly detection and predictive maintenance, making it highly suitable for improving operational efficiency in manufacturing and industrial systems. Meanwhile, MediaPipe by Google enables real-time gestures and face tracking, making it an ideal solution for human-computer interaction tasks in edge devices.

Another critical player in this space is CUBE-AI, developed by STMicroelectronics for its STM32 microcontroller family. Released in 2020, CUBE-AI optimizes machine learning models for deployment on STM32 devices, particularly in industrial IoT applications. It focuses on tasks like predictive maintenance and anomaly detection, where efficient memory management and low-power consumption are crucial. While each of these frameworks has distinct strengths and targeted use cases, developers must consider their specific application requirements when choosing the appropriate tool. Factors such as model size, performance metrics, and hardware compatibility play a significant role in determining the most suitable framework.

In the following sections, we will delve deeper into three specific frameworks: TensorFlow Lite for Microcontrollers, Edge Impulse, and CUBE-AI from STMicroelectronics, analyzing their unique contributions, use cases, and practical implementations for TinyML applications.

TensorFlow Lite is an upgraded version of TensorFlow, created by

Table 3
Overview of leading tools and frameworks in TinyML applications.

Framework, year	Developer	Applications	Performance Metrics / Model Size	Ease of Use / Deployment	Supported Hardware/ MCUs
TensorFlow Lite [100] 2017	Google Brain Team	image recognition, audio classification, and object tracking in various IoT applications.	size: ~138 KB, 30 FPS inference	High	ARM Cortex-M, ESP32, STM32, Arduino
uTensor [101] 2018	ARM	gesture recognition and audio classification for low-power wearables.	Low-latency microcontrollers	Moderate	ARM Cortex-M
Edge Impulse [102] 2019	Zach Shelby, Jan Jongboom	asset tracking and predictive maintenance	Fast real-time processing with low-power usage	High	STM32, ESP32, Arduino, Raspberry Pi
NanoEdge AI Studio [103] 2019	Cartesiam	anomaly detection and predictive maintenance	Small model footprint	Moderate	ARM Cortex, ESP32, STM32
PyTorch Mobile [104] 2019	Meta AI	computer vision and natural language processing	Efficient model inference	Moderate	Android, iOS, Linux CPUs
MediaPipe [105] 2019	Google	real-time human-computer interaction tasks	Real-time gesture tracking, low-latency performance	High	ARM Cortex-M, STM32
OpenVINO [106] 2020	Intel	object detection, facial recognition, and image segmentation.	deep learning inference	Moderate	Intel CPUs, FPGAs, GPUs
CUBE-AI [107] 2020	STMicroelectronics	predictive maintenance and anomaly detection in industrial IoT systems.	Efficient memory and deep learning	Moderate	STM32 family of microcontrollers

Google in 2017, designed for executing Machine Learning models on microcontrollers and other edge devices with constrained memory and power resources [108]. The software is compatible with several microcontroller units (MCUs) including ARM Cortex-M series, ESP32, STM, and Arduino boards. It is used with several applications, such as image classification tasks on ARM Cortex M4 [109], Oder classifications based on metal oxide sensors [110], Speech recognition [111], Driver behavior [112], and Gesture detection [113].

Fig. 5 illustrates the development timeline of TensorFlow Lite from its inception in 2017 to its various applications and advancements up until 2023. TensorFlow Lite was created by Google in 2017, specifically designed to run machine learning models on resource-constrained devices like microcontrollers and edge devices. In 2018, support for ARM Cortex-M series was introduced, allowing developers to deploy TensorFlow Lite models on ARM's widely used embedded systems. By 2019, TensorFlow Lite expanded its support to platforms such as ESP32, STM, and Arduino, further increasing its versatility for IoT applications. The year 2020 marked a significant step forward, as image classification using ARM Cortex M4 was integrated, demonstrating TensorFlow Lite's ability to handle sophisticated vision-based tasks on low-power devices. In 2021, it was used for odor classification based on metal oxide sensors, broadening its applications into environmental monitoring and smart sensing. By 2022, speech recognition capabilities were added, enabling real-time voice interactions on microcontrollers. Most recently, in 2023, TensorFlow Lite has been applied in advanced use cases such as driver behavior detection and gesture recognition, continuing to evolve and expand its role in enabling smart, AI-powered solutions on edge devices.

Edge Impulse is a complete development platform for TinyML [114]. It provides tools for data collection, model training, and deployment directly to microcontrollers. It supports several types of microcontrollers such as Arduino, Raspberry Pi, ESP32, and STM32. The platform is user-friendly and seamlessly integrates with TensorFlow Lite for Microcontrollers (MCUs). Researchers and developers have utilized this platform for a wide range of TinyML applications, such as image classification [115], voice recognition [116], and battery life cycle prediction [117].

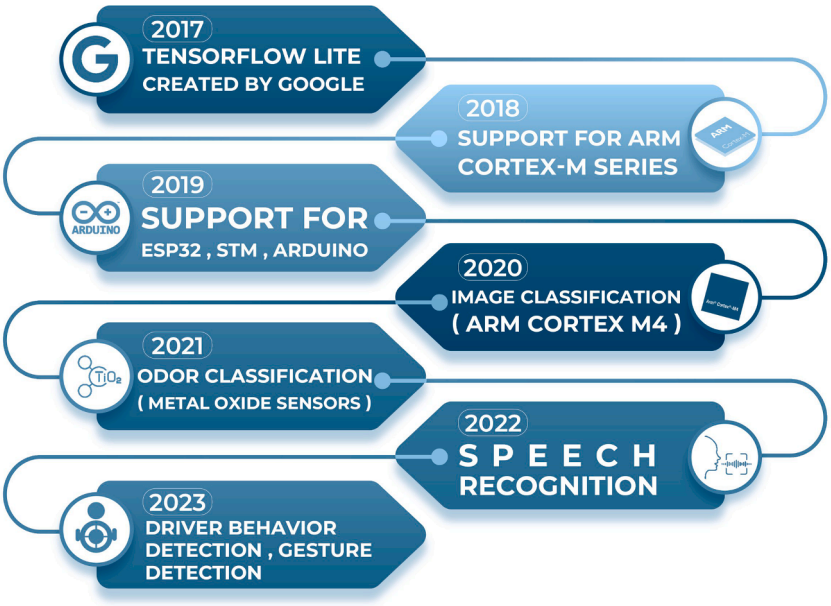


Fig. 5. Timeline of TensorFlow Lite Development and Key Applications (2017–2023).

CUBE-AI from STMicroelectronics is specifically intended for implementing artificial intelligence on STM32 Microcontrollers and is considered a component of STM32Cube [118]. This tool enables the transformation of pre-trained neural networks into C code that can be executed on STM32 devices. It is utilized in various TinyML applications, including the identification of movement patterns [119], Human activity recognition [120] and environmental predictions [121].

Table 4 presents a comparison of various studies that introduced Tiny FL frameworks for edge devices. The results indicate the presence of certain limitations that require more research and development.

4.1. Applications and challenges of FL and TinyML in IoT edge devices

The integration of Federated Learning (FL) and TinyML in IoT edge devices has shown promising applications across various sectors, including healthcare, smart cities, industrial automation, and environmental monitoring. These technologies offer significant advantages in terms of data processing, privacy, and system efficiency, although they also present certain challenges. In the healthcare sector, TinyML and FL have been utilized to enhance patient monitoring systems. For instance, FL enables the training of machine learning models on decentralized data sources, such as wearable devices, without compromising patient privacy. This approach allows for real-time health monitoring and personalized healthcare solutions while ensuring data security and compliance with privacy regulations [127,128]. The use of TinyML in these devices further optimizes power consumption and processing speed, making it feasible to deploy sophisticated models on resource-constrained devices [129]. Smart cities benefit from the integration of FL and TinyML through improved data management and decision-making processes. These technologies facilitate the deployment of intelligent systems for traffic management, energy consumption optimization, and public safety monitoring [130]. By processing data locally on edge devices, cities can reduce latency and bandwidth usage, leading to more responsive and efficient urban management systems [131,132]. In industrial automation, FL and TinyML are applied to predictive maintenance and quality control. By leveraging FL, industries can train models on data collected from various machines without transferring sensitive information to a central server, thus maintaining data privacy and security

TinyML enables the deployment of these models on edge devices, allowing for real-time anomaly detection and system optimization [133,134]. This integration not only enhances operational efficiency but also reduces downtime and maintenance costs. Environmental monitoring is another area where these technologies have made a significant impact. FL and TinyML are used to monitor air quality, water levels, and wildlife habitats by processing data locally on sensor networks. This approach minimizes data transmission costs and enhances the timeliness of environmental assessments. The ability to update models without centralized data collection also addresses privacy concerns related to sensitive environmental data [135]. Overall, the integration of FL and TinyML in IoT edge devices offers substantial benefits in terms of privacy, efficiency, and real-time data processing across various sectors. A more detailed discussion of the cross-cutting technical challenges and future research directions is provided in Section 6.

5. Lossless data compression techniques for microcontroller

Lossless data compression techniques are a must for microcontrollers, especially in cases where memory and processing power are limited, but data conservation is imperative. By efficiently compressing data without any loss of information, lossless compression techniques enhance the feasibility and scalability of TinyML and FL deployments on IoT edge devices. Here are several common techniques used for lossless data compression in microcontrollers:

- Run-Length Encoding (RLE): This method of simple data compression is very efficient for data that has a lot of such byte sequences as the same repeating bytes occurring in a row. It is performed by storing the length of consecutive blocks of data in place of the actual content. This is optimal for basic microcontroller projects, where there is a lot of redundant data [136].
- Huffman Coding: This method is based on a variable-length code table that is used to encode a source symbol like an individual byte in a file. The VLC table is designed to be of variable length, and these lengths are based on the probability of the individual source symbols. Huffman coding is a widely used method that reduces the average code length of messages [137].

  Lempel-Ziv-Welch (LZW): The LZW algorithm is a compression method that uses a dictionary of input sequences which is created as the data is processed. In data processing, sequences of repetitive

Table 4
Summary of Federated Learning Frameworks and Algorithms: Communication, Memory, Battery Life, Compatibility, and Scalability on Edge Devices.

Author, year	Framework	Algorithm	Edge devices	Communication cost	Memory	Battery life	compatibility	Scalability
name, year [17]	TinyFedTL	FedAvg	Arduino Nano 33	High	High	Long	Yes	Yes
[95]	Flower Framework	FedAvg	Mobile Phone, Nvidia Jetson	Moderate	Moderate	Short	Limited for MCUs	Yes
[122]	SFL framework	SFL	Arduino	High	Moderate	Moderate	Yes	Limited
[123]	FedTiny	FedAvg	–	Low	Low	Long	Yes	Limited (tested on specific models)
[17]	TinyReptile	FedAvg	-Raspberry Pi -Arduino Nano BLE 33 MCU	High	Moderate	Moderate	Yes	Limited
[124]	CE-FedAvg	FedAvg	-Raspberry Pi (RPi) 2B -Raspberry Pi (RPi) 3B	Low	Moderate	Short	Limited for MCUs	Yes
[97]	TensorFlow Lite	FedAvg	Arduino Nano 33 BLE	High	Low	Long	Yes	Limited
[125]	Over-the-Air FL	FedAvg	–	Low (with perfect CSI)	Low	Long	Yes	Moderate (CSI requirements)
[126]	FedPrune	FedPrune	-Raspberry Pi (RPi)	Moderate	High	Short	Limited	Yes
[21]	TensorFlow Lite	TFL-BC	–	High	Moderate	Moderate	Yes	Limited
[96]	FedAvg with Transfer Learning	FedAvg	-Arduino -ESP8266 -ESP32 -Rasp. Pi	High	High	Short	Yes	Limited

sequences are replaced with shortcodes that point to the dictionary. This is a powerful technique in scenarios where data show a high number of repetitions [137].

- Adaptive Huffman Coding: Adaptive Huffman coding keeps the Huffman tree up-to-date while the data is being transferred, thus making it suitable for situations where data attributes are unknown or vary, unlike the conventional Huffman coding, which requires prior knowledge of data statistics to be carried out [135].
- Arithmetic Coding: This is a much higher level of entropy coding than Huffman coding. It represents the whole information. The message is condensed into a denominator and numerator which is between 0 and 1. Arithmetic coding is better than Huffman coding in the compression of data, particularly with symbols of low probability and when high precision is necessary [138].
- Delta Encoding: Delta encoding is mostly used for data with minor differences between successive data points. It encodes the difference between consecutive data points rather than the data points themselves. This can be particularly helpful in scenarios such as sensor data monitoring, where the differences between subsequent measurements are typically minimal [139].
- Entropy Encoding: General category encompassing techniques such as Huffman and Arithmetic coding. It encodes data based on variations in the frequency of data item occurrences. This method is essential for reducing data into the most minimal number of bits [140].

Using lossless compression techniques with edge devices has many benefits, the main one being a decrease in power and communication costs. An IoT device gathers data from several sensors, encodes it, compresses it, and temporarily stores it in memory. Eventually, it uses multiple IoT protocols to transmit the compressed data in a single packet to the gateway, as illustrated in Fig. 6. Because it doesn't require direct data transmission after collection, this method enables Internet of Things devices to operate on batteries for extended periods of time.

Table 5 presents a comparison of several paper contributions focusing on the application of lossless compression algorithms on IoT edge devices, particularly in terms of the compression ratio. While the results demonstrate variations in the quantity of data gathered and compressed before being stored in the microcontroller's memory, the highest compression ratio does not necessarily indicate the "best" algorithm.

For example, the MDEH algorithm achieves the highest compression ratio at 93.7 %, suggesting it is highly efficient in reducing data size. However, other factors such as energy efficiency, processing time, and resource requirements must also be considered when evaluating its practicality for IoT edge devices. In some use cases, algorithms with lower compression ratios, such as FLAC or VLDE, might be more suitable due to their lower computational overhead and power consumption, which are critical in resource-constrained environments like IoT sensor nodes.

Table 5
Comparison of Lossless Data Compression Techniques in IoT edge devices.

Author and Year [Ref.]	Application	Algorithm	Compression Ratio
[141], 2021	wireless EMG sensors.	FLAC VLDE	28 % 42 %
[142], 2021	IoT devices in cloud environments.	S-LZE Z-LEC	51 % 27 %
[143], 2021	deep learning-based IoT networks.	Adaptive Huffman algorithm	55–68 %
[144], 2021	IoT sensor nodes in smart agriculture,	ATP MDE MDEH	83 % 87.8 % 93.7 %
[145], 2019	wireless sensor networks.	NIS	89.13 %

Thus, while MDEH demonstrates excellent performance in terms of compression ratio, it is not necessarily the optimal solution across all IoT scenarios. Each compression algorithm must be assessed based on the specific requirements of the application, including energy efficiency, computational complexity, and the nature of the data being compressed.

6. Challenges and future directions

Based on the outcomes of this assessment and the literature review, the convergence of Federated Learning (FL), TinyML, and the Internet of Things (IoT) presents significant potential for advancing Edge AI. Despite the promise these technologies hold, there remains a notable absence of comprehensive research and practical implementations that effectively integrate all three. Current research predominantly focuses on one or two technologies at a time, offering mostly theoretical frameworks or isolated implementations rather than cohesive, unified solutions. In particular, the existing literature on federated learning implementations on microcontrollers is constrained by several challenges, with high power consumption and communication overhead being among the most critical. Furthermore, the limited memory capacity of microcontrollers poses additional hurdles, preventing the retention of data for extended periods before transmission to Gateway devices, a necessity in many IoT applications.

Another major challenge is the heterogeneity of data generated by IoT devices operating in diverse environments. This variability complicates model convergence and reduces accuracy in FL systems, as illustrated in Fig. 7. Moreover, these factors create significant barriers to

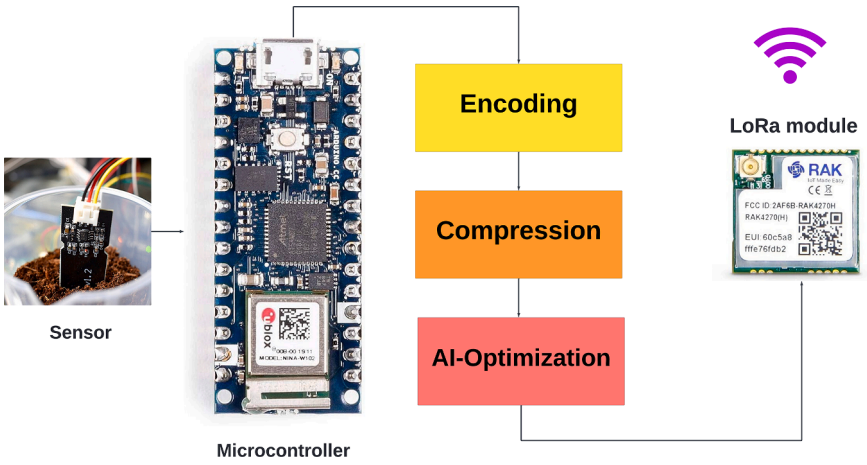


Fig. 6. Data compression proposed approach for IoT devices.

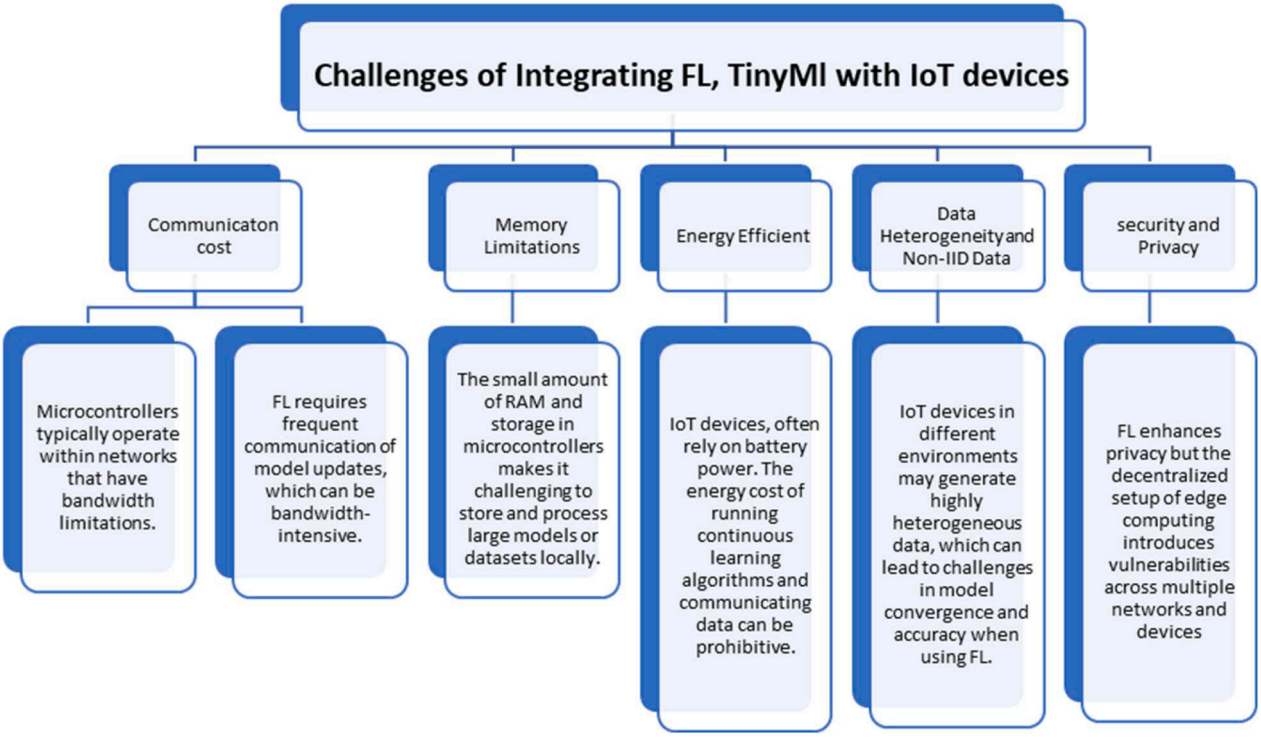


Fig. 7. Several challenges of using FL and TinyML on edevices.

powering IoT nodes via batteries, which are crucial for many edge-based applications. The confluence of these issues has limited the large-scale deployment of autonomous IoT systems. These identified gaps highlight a promising area for future research, ripe with opportunities for innovation. The current lack of integrated strategies that effectively combine FL, TinyML, and IoT technologies underscores a key challenge in the field: developing practical, scalable, and privacy-preserving solutions that can be implemented in real-world environments. Bridging this gap between theory and practical application is critical to unlocking the full potential of these technologies.

With the rapid expansion of the IoT market and the growing demand for intelligent, autonomous devices, there is an urgent need for networking solutions that capitalize on the combined strengths of FL and TinyML. Future research must prioritize the development of intelligent frameworks that address these challenges and fully exploit the potential of Edge AI, thus enabling the next generation of IoT applications designed for real-world scalability and impact.

6.1. Research gap

While this proposed framework offers a comprehensive approach to integrating Federated Learning (FL), TinyML, and the Internet of Things (IoT) for edge-based machine learning, several research gaps remain that need to be addressed to fully unlock the potential of these technologies:

- Handling Data Heterogeneity and Non-IID Data: As identified in prior sections, IoT devices often generate heterogeneous data due to differences in their operating environments. This variability poses a significant challenge in ensuring model convergence and accuracy when using FL. Current methods still struggle to manage non-IID data effectively in real-world settings, where environmental and usage conditions can vary significantly across IoT nodes. Addressing this issue will require more sophisticated algorithms capable of balancing diverse data inputs while maintaining robust and accurate model updates.
- Memory and Power Constraints: While advancements in TinyML have made it possible to run machine learning models on microcontrollers, the limited memory capacity and energy efficiency of these devices continue to impose significant restrictions. Existing models often exceed the available memory or require more power than is feasible for battery-operated devices. Further research is necessary to create lightweight, power-efficient models that can perform well on devices with constrained resources.
- High Communication Costs: The proposed framework addresses communication efficiency through data compression, but frequent model updates in FL, especially in large networks, can still lead to high communication overhead. Although model pruning and update aggregation techniques have shown promise, more work is needed to develop communication protocols that further reduce bandwidth consumption without sacrificing model accuracy or performance.
- Scalability and Practical Deployments: While FL offers a privacy-preserving solution for distributed machine learning, its practical implementation in large-scale IoT networks remains largely theoretical. Existing research has yet to demonstrate the deployment of FL, TinyML, and IoT in a scalable and operational framework that can handle the demands of real-world applications. Key areas that require further exploration include managing massive IoT networks, ensuring synchronization across nodes, and dealing with network disruptions.
- Security and Privacy: Although FL addresses some privacy concerns by keeping data localized on the device, the decentralized nature of the system introduces new security

vulnerabilities. Malicious attacks, model poisoning, and Unauthorized access to the aggregated models remain potential threats. More research is needed to develop robust security protocols and mechanisms for ensuring trust in FL systems deployed in dynamic IoT environments. These research gaps offer significant opportunities for innovation and improvement in the field. By addressing these challenges, future work can further enhance the efficiency, scalability, and security of systems

that integrate FL, TinyML, and IoT, paving the way for smarter and more autonomous IoT ecosystems.

6.2. Innovative solution

Following a thorough analysis of the research gaps and a detailed review of the existing literature, it is evident that the current frameworks for integrating Federated Learning (FL), TinyML, and the Internet of Things (IoT) fall short of providing practical, scalable, and efficient solutions. To address these challenges and capitalize on the potential of these technologies, a novel framework that integrates these three domains is proposed in this study to optimize the deployment of machine learning algorithms on microcontrollers in IoT ecosystems.

This framework aims to overcome key limitations such as high communication costs, memory constraints, and data heterogeneity, all of which have been identified as significant obstacles to advancing Edge AI. The proposed solution is not only focused on addressing these technical challenges but also emphasizes privacy preservation and scalability, enabling real-world applications to thrive.

Central to this framework is an efficient mechanism that facilitates direct, on-device machine learning across a distributed network of IoT nodes, all connected via a LoRa network. The process begins at the IoT nodes, each equipped with sensors to collect environmental or application-specific data (as shown in Fig. 8). This data is temporarily stored locally and undergoes compression to optimize transmission efficiency. Once compressed, the data is sent to a local server, implemented on a Raspberry Pi, which serves as an intermediary hub. The server aggregates the data from various nodes and participates in a larger FL network, ensuring global collaboration on model training while conserving bandwidth.

A distinctive feature of this framework is its dual focus:

- **First**, it leverages local data processing to significantly reduce both latency and bandwidth consumption.
- **Second**, it utilizes Federated Learning to keep AI models updated without compromising the privacy of individual IoT nodes. The framework ensures that each IoT node is equipped

with the most up-to-date models, which enhances the overall efficiency and capability of the IoT network.

By integrating FL, TinyML, and IoT, this framework offers a scalable, privacy-preserving, and energy-efficient solution for deploying advanced machine learning directly at the edge. It not only addresses the key challenges of power consumption, memory limitations, and data heterogeneity but also enables IoT nodes to operate more intelligently, autonomously, and efficiently. This innovation provides a pathway for smarter, more responsive IoT ecosystems, designed to meet the demands of real-time decision-making while making optimal use of available network resources.

Evaluation Metrics:

To rigorously evaluate the proposed FL-IoT framework, we define the following key performance metrics:

- Accuracy: Evaluated using metrics such as classification accuracy, R^2 score, or F1-score to assess predictive model performance.
- Communication Cost: Quantified by measuring the number of transmitted bits or communication rounds required per federated learning cycle.
- Energy Consumption: Assessed using milliwatts (mW) or Joules per training iteration, leveraging hardware-based power monitoring tools.
- Memory Footprint: Determined by measuring model size and storage requirements in kilobytes (KB), reflecting practicality for deployment on resource-constrained IoT microcontrollers.

These metrics provide a clear and comprehensive performance assessment, demonstrating the framework's suitability and efficiency for IoT edge applications.

7. Conclusion

The integration of Federated Learning (FL), TinyML, and IoT within resource-constrained edge devices represents a significant advancement in the field of intelligent computing. This paper provides a comprehensive examination of the strategies, challenges, and opportunities associated with this innovative convergence. By focusing on the technical aspects and broader implications, we have highlighted the critical role of edge computing in enhancing data processing and analytics capabilities directly at the device level. Our study stands as a pioneering effort to combine FL with TinyML, specifically tailored to improve learning capabilities in compact environments such as microcontrollers. We have identified key challenges, including communication costs, memory and processing limitations, and data security concerns, which must be addressed to optimize model efficiency on IoT edge devices. Moreover, we have pointed out significant gaps in existing research, particularly the lack of comprehensive studies and practical applications that fully integrate FL, TinyML, and IoT technologies. By suggesting strategies for frequent AI model updates on IoT devices, our study encourages the development of smarter microcontroller systems, thereby reducing power consumption and communication expenses. The proposed future research directions aim to overcome current obstacles and enhance the deployment of AI models on microcontrollers, paving the way for new technological capabilities in intelligent, decentralized, and privacy-preserving computing. In conclusion, this paper serves as a

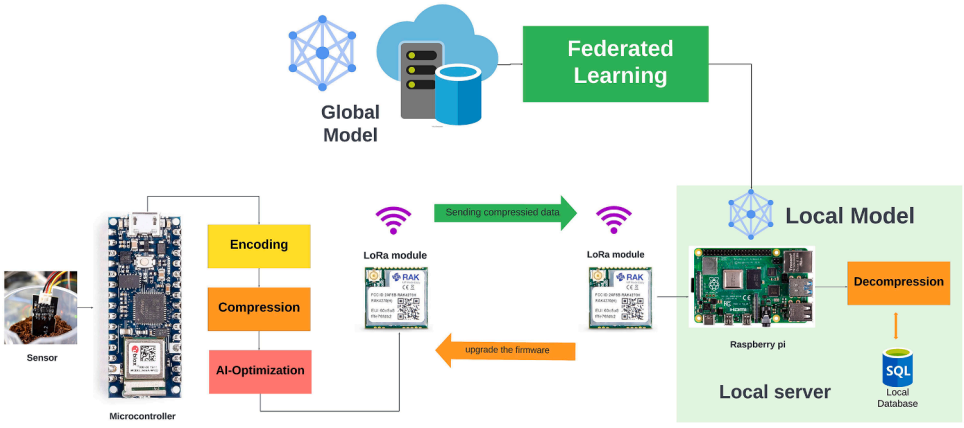


Fig. 8. The proposed IoT FL framework overviews.

catalyst for further research and development in this dynamic field, inspiring innovative solutions that will drive the next generation of IoT applications. By addressing the challenges outlined, we can unlock the full potential of these technologies, creating more intelligent and efficient systems that benefit various sectors and industries.

CRediT authorship contribution statement

Montaser N.A. Ramadan: Writing – review & editing, Validation, Methodology, Conceptualization. **Mohammed A.H. Ali:** Writing – review & editing, Supervision. **Shin Yee Khoo:** Writing – review & editing. **Mohammad Alkhedher:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

ADU author acknowledges financial support from Abu Dhabi University's Office of Research and Sponsored Programs.

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